Solution Architecture

AbbVie R&D Convergence Hub (ARCH)

Aligning Knowledge Integration with Pharmaceutical R&D Excellence

24-Month Strategic Implementation Roadmap

Enterprise Knowledge Accelerator (EKA)

Knowledge Integration & Articulation (KIA)

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Executive Summary

This Solution Architecture defines the technical blueprint and 24-month implementation roadmap for the AbbVie R&D Convergence Hub (ARCH), a unified knowledge platform engineered to transform AbbVie's research enterprise from a data-driven to a knowledge-driven paradigm. ARCH integrates a harmonized 124TB data corpus, a labeled property graph containing over 30 million nodes and nearly 2 billion relationships, and a federated AI layer to deliver traceable, hallucination-free insight at the point of scientific decision-making.

The platform addresses a competitive mandate: as AbbVie expands its footprint in high-complexity therapeutic areas—Oncology, Immunology, and Neuroscience—the ability to validate a target's causal role in disease determines the difference between a multi-billion-dollar success and a clinical failure. ARCH compresses discovery timelines from months to minutes, reduces late-stage attrition, and delivers quantified business value across the entire drug development lifecycle.

Strategic Pillars

- **Knowledge Democratization:** Serving as the trusted knowledge partner for R&D and Corporate Strategy, ensuring decision-making is fueled by shared intelligence rather than fragmented silos.
- **AI-Enabled Insight:** Powering enterprise AI platforms by integrating published external literature with formalized internal insights to drive high-velocity discovery.
- **Information Unlocking:** Leveraging patented NLP and machine learning to extract actionable relationships from a harmonized global corpus, unlocking information that makes cures possible.

Document Scope

This document serves as the authoritative reference for the ARCH solution architecture. It encompasses the platform's layered technical architecture, a phased 24-month implementation roadmap with explicit deliverables and success criteria, advanced clinical capabilities including Patient Digital Twins and Digital Health Technology integration, data governance standards anchored in FAIR Principles and MBSE methodology, specialized user experience platforms, semantic governance frameworks, and strategic risk mitigation strategies.

Layered Solution Architecture

The ARCH platform is engineered as a four-layer convergence architecture, each layer building upon the previous to transform raw data into actionable, traceable knowledge. This "Architecture of Truth" utilizes a bi-layered data and knowledge approach, breaking the organizational silos that traditionally separate medical affairs, early discovery, and clinical research.

Layer 1: Data Foundation & Ingestion

The foundational layer manages the harmonized data corpus—a massive 124TB environment connecting 200+ internal and external data sources. This layer is governed by strict FAIR Principles (Findability, Accessibility, Interoperability, Reusability) and enforces unyielding Service Level Agreements (SLAs) for all ingested data assets.

Core Technical Components

- **Data Movement & Automation (Modak Nabu):** Manages underlying data flow, automated ingestion pipelines, and ETL orchestration at enterprise scale. All ingestion workflows are designed using Model-Based Systems Engineering (MBSE) principles to guarantee zero data loss during integration from legacy systems and disparate source schemas.
- **Named Entity Recognition (SciBite TERMite):** Normalizes biomedical concepts—genes, proteins, chemicals, diseases—with high precision, achieving "PhD accuracy at Big Data scale" through synonym mapping to a single ontological concept.
- **Ontology Management (SciBite CENtree):** Democratizes the creation and management of enterprise-wide controlled vocabularies, providing the semantic backbone that transforms disparate strings into unified biological things.
- **FAIR Data Quality Gateway:** Every data asset entering ARCH must pass through a quality gateway that validates compliance with FAIR Principles. Automated quality scoring assigns Findability, Accessibility, Interoperability, and Reusability metrics before data is promoted to the knowledge layer.

Layer 2: Semantic Harmonization & Knowledge Graph

This layer transforms normalized data into structured, semantically enriched knowledge. The AbbVie Repository of Knowledge (ARK) is a labeled property graph managing over 30 million nodes and nearly 2 billion relationships, providing a comprehensive map of biomedical cause-and-effect.

Core Technical Components

- **Relationship Extraction (Tellic):** Extracts over 1 billion biomedical relationships from the global literature corpus, transitioning from statistical co-occurrence to relationship-theme extraction using distributional semantics. The ARK graph is populated with 80+ distinct relationship types (e.g., inhibits, agonism, upregulates) to model complex biological interactions.
- **Neo4j Semantic Engine:** Serves as the graph database engine, enabling Cypher-based querying of the knowledge graph with full provenance tracing. Every triple is tagged with source, timestamp, and confidence metadata.

- **Fidelity Threshold Enforcement:** A high threshold value (0.9) is applied to all NLP-derived triples. Insights from the GNBR-Nexus evaluation confirm that applying high thresholds to specific relationship classes like “inhibits” yields high-fidelity triples not reported in structured datasets, ensuring the Knowledge Graph remains an Architecture of Truth.
- **Entity Resolution & the Reveal-Reframe-Resist Model:** A governance layer that identifies NLP failures (e.g., confusing COX-1 abbreviations), ensures context-driven mapping to precise concept IDs in ontologies like UniProt, and uses expert curation combined with LLM grounding to resist the incorporation of mislabeled entities.

Layer 3: AI & Prompt Engineering (Semantic Gateway)

The AI layer operationalizes the Context Engine, moving beyond simple keyword matching to deliver relevancy grounded in ontological metadata. This layer acts as the Semantic Gateway between the user and the knowledge graph, ensuring every AI interaction is contextually precise.

Core Technical Components

- **AWS Bedrock Router:** Routes user queries to the appropriate language model based on persona, permissions, and therapeutic domain context. Identifies whether the user is a Scientist, Practitioner, or Patient and applies the corresponding language model and permission set.
- **Enterprise Prompt Library:** A curated library of ontology-enriched prompt templates that wraps user requests in structured metadata—the 5 Ws (Who, What, Where, When, Why)—before the query reaches the LLM. This ensures intent-driven prompting rather than raw prompt submission.
- **Federated RAG 2.0 Architecture:** Utilizes domain-specific vector databases as virtual Specialized Language Models (SLMs), querying the Neo4j graph directly to ground LLM outputs in verified semantic data. Cypher query generation eliminates hallucinations and provides traceable provenance for every answer.
- **Context Engine (5 Pillars of Relevancy):** Maps every query through five ontological dimensions: Who (Party/Persona), What (Product/Asset), Where (Location/Domain), When (Time/Event), and Why (Issue/Service/Intent). This ensures that identical queries from different personas return contextually appropriate results.

Layer 4: User Interface & Application Delivery

The presentation layer delivers intuitive, task-tailored applications for every stakeholder persona through the Unity design system (TXI Digital). This layer provides 18 distinct views of a molecule or gene target, reducing time-to-insight from months to minutes.

Platform Portfolio

- **ARCH Search:** The primary search interface enabling natural-language querying of the knowledge graph, with results grounded in ARK data and full provenance tracing.
- **Target Dossier:** A comprehensive view aggregating pathways, variants, phenotypes, and safety signals for a specific biological target, replacing months of manual curation with a unified, real-time interface.

- **Safety Dossier:** Aggregates adverse event data, toxicology findings, and safety signals across internal clinical trials and external literature, rendered as an auditable, GxP-compliant view.
- **AbbVie Science Oncology Platform:** A modernized, specialized interface delivering tailored research updates directly to healthcare practitioners in oncology. This platform extends the ARCH ecosystem beyond internal R&D, providing practitioners with curated, graph-grounded insights on emerging targets, clinical trial updates, and translational research relevant to their therapeutic focus.
- **Rare Hopes UX:** A purpose-built patient-and-clinician interface designed to translate complex graph-derived hypotheses into actionable insights for rare diseases. Rare Hopes transforms the dense biomedical relationships in ARK into accessible language for patients and their doctors, enabling informed shared decision-making for conditions where traditional evidence bases are sparse.
- **Ask ARCH:** A natural language conversational interface for knowledge graph querying, grounding LLM responses in structured ARK data using vector search and Cypher query generation.

Core Technology Integration Matrix

The following matrix maps each core technology component to its functional role, strategic partner, and architectural layer within the ARCH platform.

Component	Functional Role	Strategic Partner	Architecture Layer
Data Movement & Automation	Automated ingestion, ETL orchestration, MBSE-governed data flow	Modak Nabu	Layer 1: Foundation
Ontology Management	Enterprise-wide controlled vocabularies and semantic governance	SciBite CENtree	Layer 1: Foundation
Named Entity Recognition	Biomedical concept normalization with PhD-level precision	SciBite TERMite	Layers 1–2
Relationship Extraction	1B+ biomedical relationship extraction via distributional semantics	Tellic	Layer 2: Semantic
Graph Database Engine	Labeled property graph with Cypher querying and provenance tracing	Neo4j	Layer 2: Semantic
AI Orchestration & Routing	Persona-based LLM routing, Federated RAG 2.0, prompt engineering	AWS Bedrock	Layer 3: AI
UI/UX Design System	Intuitive, task-tailored applications for scientists and practitioners	TXI Digital (Unity)	Layer 4: Interface

24-Month Strategic Implementation Roadmap

The implementation follows a "quick but not cheap" mandate, structured across three phases over a precise 24-month execution timeline. Each phase builds upon the previous, delivering incremental business value while progressively advancing the platform's capabilities from foundational data ingestion through autonomous agentic intelligence.

Phase 1: Foundation (Months 1–6)

Objective: Establish the unified data foundation, achieve semantic normalization at scale, and deliver the initial knowledge graph with verified, high-fidelity relationships.

Stage I: Data Ingestion & Normalization

- **Primary Deliverable:** A unified data lake connecting 200+ internal and external sources with full FAIR compliance.
- **MBSE-Governed Ingestion:** All data integration workflows are designed and validated using Model-Based Systems Engineering principles, guaranteeing zero data loss during migration from legacy systems and disparate source schemas. MBSE serves as the primary methodology for structurally sound integration, preventing the accumulation of integration debt.
- **FAIR Data Quality SLAs:** Every ingested data asset must satisfy unyielding Service Level Agreements across all four FAIR dimensions. Automated quality scoring gates prevent non-compliant data from entering the ARCH ecosystem. Findability mandates require persistent identifiers and rich metadata; Accessibility requires standardized retrieval protocols; Interoperability enforces controlled vocabulary mapping; Reusability demands clear provenance and usage licensing.
- **NER & Synonym Mapping:** Deployment of SciBite TERMite for biomedical concept normalization, mapping all entities to single ontological concepts via SciBite CENtree vocabularies.
- **Clinical Data De-identification:** De-identification pipelines for data from over 2,000 clinical trials, maintaining ethical exclusions for pediatric and prison populations, enabling risk-based data democratization for secondary use.
- **Measure for Success:** Achieving PhD accuracy at Big Data scale through NER and synonym mapping; 100% FAIR compliance scoring on all promoted data assets; zero data loss validated through MBSE traceability audits.

Stage II: Knowledge Extraction & Modeling

- **Primary Deliverable:** Population of the ARK labeled property graph with semantically enriched, high-fidelity biomedical relationships.
- **Distributional Semantics Extraction:** Transition from statistical co-occurrence to relationship-theme extraction using Tellic's NLP engine, populating the graph with 80+ distinct relationship types modeling complex biological interactions.
- **Fidelity Threshold Enforcement:** Application of a 0.9 confidence threshold for all NLP-derived triples, validated against GNBR-Nexus benchmarks to ensure only high-fidelity relationships enter the graph.

- **Entity Resolution Governance:** Deployment of the Reveal-Reframe-Resist model for entity resolution, with expert curation workflows to identify and correct NLP misclassifications before graph population.
- **Measure for Success:** ARK graph populated with 30M+ nodes and initial relationship sets; all triples above 0.9 fidelity threshold; entity resolution accuracy validated by domain scientists.

Phase 2: Acceleration (Months 6–12)

Objective: Deploy user-facing applications, launch specialized UX platforms, introduce Patient Digital Twin capabilities, and establish the AI-powered Semantic Gateway for context-aware querying.

Stage III: Interface & Application Deployment

- **Primary Deliverable:** Full deployment of ARCH Search, Target and Safety Dossiers, AbbVie Science Oncology Platform, and Rare Hopes UX via the Unity design system.
- **Target & Safety Dossiers:** Delivery of 18 distinct views of a molecule or gene target, aggregating pathways, variants, phenotypes, and safety signals into a unified, GxP-compliant interface. Scientists transition from months of manual curation to minutes of graph-powered insight.
- **AbbVie Science Oncology Platform:** Launch of the modernized oncology interface delivering tailored research updates, emerging target intelligence, and clinical trial findings directly to healthcare practitioners. This platform extends ARCH's value beyond internal R&D into the practitioner ecosystem, providing curated graph-grounded insights for oncology decision-making.
- **Rare Hopes UX:** Deployment of the specialized patient-and-clinician interface for rare diseases. Rare Hopes translates complex graph-derived hypotheses from ARK into accessible, actionable insights for patients and their doctors, enabling shared decision-making where traditional evidence bases are sparse. The platform surfaces potential therapeutic connections that would otherwise remain buried in research literature.
- **Measure for Success:** 18 dossier views operational; time-to-insight reduced from months to minutes; Rare Hopes UX validated with rare disease patient advisory boards; AbbVie Science Oncology Platform active with practitioner user base.

Patient Digital Twin Capabilities

- **Primary Deliverable:** Initial deployment of Patient Digital Twin infrastructure leveraging observational data and historical control arms.
- **Observational Data Integration:** Integration of real-world patient data, historical clinical trial outcomes, and biomarker profiles into patient-specific computational models. These Digital Twins simulate placebo outcomes for specific patient profiles, enabling researchers to predict individual responses to therapeutic interventions before physical trials begin.
- **Clinical Trial Optimization:** Digital Twin simulations are used to reduce redundant experiments by modeling expected outcomes across patient subpopulations, optimizing clinical trial sample sizes, and identifying optimal patient stratification strategies. This capability directly reduces clinical trial costs while improving statistical power.
- **Measure for Success:** Digital Twin models validated against historical control arm data with predefined concordance thresholds; demonstrated reduction in proposed trial sample sizes for at least one active program.

Context Engine & Semantic Gateway Deployment

- **Primary Deliverable:** Operational deployment of the AWS Bedrock Router, Enterprise Prompt Library, and Context Engine with full 5 Pillars of Relevancy integration.

- **Persona-Based Query Routing:** The Bedrock Router identifies user persona (Scientist, Practitioner, Patient) and applies corresponding permissions, language models, and prompt templates automatically.
- **Intent-Driven Prompting:** The Enterprise Prompt Library wraps every user request in ontological metadata (Who, What, Where, When, Why) before it reaches the LLM, ensuring that identical queries from different personas return contextually appropriate results.
- **Cypher Query Generation:** The Semantic Gateway translates enriched prompts into precise Cypher queries against the ARK graph, grounding every LLM response in verified data with full provenance tracing.
- **Measure for Success:** Context Engine resolving polymorphic meaning correctly across all three persona types; LLM responses grounded with traceable provenance for 100% of graph-sourced answers.

Phase 3: Evolution (Months 12–24)

Objective: Advance to autonomous agentic capabilities, integrate Digital Health Technology endpoints, scale Patient Digital Twins, and deploy the Gaia Platform for continuous recursive learning.

Stage IV: Generative AI & Agentic Integration

- **Primary Deliverable:** Ask ARCH conversational interface, Large Action Models (LAMs), and the Gaia Platform for autonomous knowledge evolution.
- **Ask ARCH:** Full deployment of the natural language conversational interface for knowledge graph querying. Users interact with ARCH through natural language; the system generates Cypher queries, retrieves grounded results, and articulates answers with full provenance and confidence scoring.
- **Large Action Models (LAMs):** Evolution from retrieval-only to action-oriented AI agents. LAMs autonomously execute multi-step R&D workflows—from hypothesis generation through literature validation to experimental design recommendation—mapping action model workflows to specific R&D milestones.
- **Gaia Platform:** The recursive learning platform where the ontology does not just inform but executes. Gaia enables continuous, autonomous enrichment of the knowledge graph through feedback loops from experimental outcomes, clinical trial results, and real-world evidence. The graph evolves as new knowledge is generated, maintaining its status as a living Architecture of Truth.
- **Measure for Success:** Ask ARCH handling complex multi-hop queries with >95% provenance accuracy; LAMs autonomously completing defined R&D workflow sequences; Gaia Platform demonstrating measurable graph enrichment from recursive learning cycles.

Digital Health Technology & Wearables Integration

- **Primary Deliverable:** Integration of Digital Health Technology (DHT) endpoints, including wearable device data streams, into the ARCH knowledge ecosystem.
- **Wearable Data Ingestion:** Integration of continuous physiological data from wearable devices (smartwatches, biosensors, activity trackers) into the ARCH data foundation. DHT endpoints capture objective, real-world patient outcomes—including activity levels, heart rate variability, sleep patterns, and movement biomarkers—providing continuous longitudinal data that complements traditional clinical endpoints.
- **Patient Digital Twin Maturation:** Wearable data streams feed directly into Patient Digital Twin models, enriching simulations with real-time physiological signals. This enables dynamic, continuously updated patient models that evolve beyond static historical data, providing researchers with living computational representations of patient health trajectories.
- **Real-World Evidence Generation:** DHT integration supports post-market surveillance and real-world evidence generation by correlating wearable-derived biomarkers with therapeutic outcomes, creating a closed-loop feedback system between clinical deployment and knowledge graph enrichment.
- **Measure for Success:** DHT data streams integrated and normalized within FAIR-compliant pipelines; wearable-derived endpoints incorporated into at least two active Digital Twin models; real-world evidence generation pipeline operational.

Roadmap Summary Matrix

Phase / Timeline	Stage	Key Deliverables	Success Criteria
Phase 1: Foundation (Months 1–6)	Stage I: Data Ingestion & Normalization	Unified data lake (200+ sources) MBSE-governed pipelines FAIR quality gateway	PhD accuracy at Big Data scale 100% FAIR compliance
	Stage II: Knowledge Extraction & Modeling	ARK property graph (30M+ nodes) 80+ relationship types	0.9 fidelity threshold Entity resolution validated
Phase 2: Acceleration (Months 6–12)	Stage III: Interface & Application Deployment	Target & Safety Dossiers (18 views) AbbVie Science Oncology Platform Rare Hopes UX	Time-to-insight: months → minutes Patient advisory validation
	Patient Digital Twins & Context Engine	Digital Twin infrastructure AWS Bedrock Semantic Gateway Enterprise Prompt Library	Digital Twin concordance validated 100% provenance on graph answers
Phase 3: Evolution (Months 12–24)	Stage IV: Generative AI & Agentic Integration	Ask ARCH conversational AI Large Action Models (LAMs) Gaia recursive learning platform	>95% provenance accuracy Autonomous workflow execution
	DHT, Wearables & Digital Twin Maturation	Wearable data integration Living Digital Twin models Real-world evidence pipeline	DHT FAIR-compliant pipelines RWE generation operational

Federated Semantic Governance Architecture

The ARCH knowledge platform depends on a federated semantic governance architecture that balances top-down strategic governance with bottom-up operational reality. This architecture defines where ontologies are authored and maintained, how top-down semantic structures are reconciled with bottom-up data realities, and how that reconciliation is enforced consistently across the Neo4j graph topology. By distributing governance across purpose-built platforms—Altair Graph Studio for high-level structure, SciBite CENtree for scientific depth, and Neo4j for the ground truth of assets—the architecture avoids the monolithic trap while maintaining high fidelity through AI-driven mapping. Without this governance layer, the “Architecture of Truth” collapses into another integration silo.

Four-Level Governance Topology

The ARCH semantic architecture operates as a four-level federated governance topology. Each level has a distinct authorship model, governance cadence, tooling environment, and isolation boundary. All four levels are bound together through formal mapping assertions and AI-driven alignment agents that guarantee semantic coherence from the highest level of abstraction down to the physical Neo4j node and edge labels.

Level 1: Core Unifying Ontology — The Constitutional Layer (Altair Graph Studio)

The Core Unifying Ontology constitutes the nine foundational categories—Process, Service, Asset, Product, Party, Location, Time, Event, and Issue—that serve as the stable semantic ceiling for the entire ARCH ecosystem. These are not domain-specific vocabularies; they are abstract, enterprise-wide categories that define the fundamental types of things AbbVie reasons about across all business domains. Their purpose is to guarantee that no matter how many domain ontologies or data sources are federated into ARCH, every concept can be traced upward to one of these nine root categories. Level 1 functions as the “constitutional” layer of the enterprise—intentionally slow-moving, requiring cross-functional change control, and insulated from the high-churn scientific data in CENtree or the technical metadata in Neo4j.

Authorship and Governance: The Core Unifying Ontology is authored and maintained by the Enterprise Knowledge Accelerator (EKA) team in collaboration with the Knowledge Integration and Articulation (KIA) governance council. The ontology is modeled in OWL 2 and managed within Altair Graph Studio’s governance environment, which provides formal versioning, role-based access control, SHACL constraint validation, and export to multiple serialization formats (RDF/XML, Turtle, JSON-LD). Changes to the Core Unifying Ontology follow a formal change-control process requiring cross-functional review from enterprise architecture, data governance, and domain science leadership. Release cadence is quarterly at most; the constitutional layer is intentionally slow-moving to provide a stable foundation.

Isolation Strategy: By maintaining the Core Unifying Ontology in Altair Graph Studio—separate from the scientific vocabulary management in CENtree and the physical graph topology in Neo4j—the business logic is insulated from both high-churn scientific data updates and technical schema changes. A quarterly CENtree release adding 500 new neuroscience terms does not require any change to Level 1. A Neo4j schema migration adding new property keys does not propagate upward. The constitutional layer remains stable while the layers below it evolve.

Level 2: Business Domain Ontologies — The Organizational Layer (Altair Graph Studio)

Level 2 ontologies specialize the Core Unifying Ontology for specific business units, functions, and organizational contexts. Examples include the Oncology Business Domain (specializing Product and Asset for oncology pipeline management), the Clinical Operations Domain (specializing Process and Event for trial lifecycle governance), and site-specific operational domains such as the Ludwigshafen Manufacturing Domain or Lake County R&D Domain. Each Level 2 ontology is formally aligned to the Core through rdfs:subClassOf and owl:equivalentClass assertions.

Componentization: Level 2 provides the “plug-in” architecture for organizational growth. When AbbVie acquires a new company, enters a new therapeutic area, or establishes a new site like Ludwigshafen, a new Level 2 component is created without redesigning the Core. The new component imports only the specific branches of the Core Unifying Ontology it requires, inheriting enterprise-wide semantics while maintaining domain-specific focus. This prevents organizational sprawl from degrading semantic coherence—each new business unit gets its own governed vocabulary space that is structurally anchored to the enterprise foundation.

Authorship and Governance: Level 2 ontologies are co-authored by business domain owners and knowledge engineers within Altair Graph Studio, using the same OWL 2 tooling as Level 1 but with domain-scoped access permissions. Governance follows a monthly review cadence managed by the KIA team, with automated validation checks confirming that all Level 2 concepts maintain valid subsumption paths to the Core Unifying Ontology. Level 2 ontologies are published to SciBite CENtree as reference vocabularies, making them available to downstream consumers including the NER pipeline, the domain ontology authors, and the Neo4j graph schema generator.

Level 3: Scientific Domain Ontologies — The Knowledge Warehouse (SciBite CENtree)

Level 3 represents the dense, multi-layered hierarchies of healthcare and scientific domains that require specialized vocabulary management. These are the therapeutic-area-specific vocabularies that power the NER pipeline, the relationship extraction engine, and the knowledge graph entity layer: the Oncology Target Ontology (specializing Product and Asset for molecules, biological targets, and biomarkers), the Clinical Safety Ontology (specializing Issue and Event for adverse events, toxicology endpoints, and regulatory findings), the Neuroscience Pathway Ontology (specializing Process for neural signaling cascades and mechanism-of-action modeling), and the Immunology Pathway Ontology (specializing Process for immune signaling cascades).

Why CENtree for Level 3: CENtree is purpose-built for managing exactly this type of content—dense biomedical hierarchies with cross-references to external authorities (UniProt, ChEBI, MeSH, SNOMED CT, Gene Ontology), synonym management across multiple languages and nomenclature systems, and direct integration with the SciBite TERMite NER engine. Attempting to manage these vocabularies in a general-purpose OWL editor would sacrifice the domain-specific features that make CENtree valuable: term lifecycle management (draft, review, approved, deprecated), automated diffing against external ontology releases, and collaborative web-based editing that empowers domain scientists to curate their own vocabularies.

Scale Without Clutter: Because CENtree only handles the breakdown of healthcare-specific and scientific domains, it does not get cluttered with business operations data (HR, Finance, Manufacturing Operations) which remains in Level 2. A scientist browsing the Oncology Target Ontology sees only oncology-relevant concepts—not manufacturing process terms from Ludwigshafen or financial reporting categories from Corporate.

CENtree Application Ontology Architecture

CENtree supports multiple independent ontologies and a mechanism for creating Application Ontologies—derivative versions that manage combinations or subsets of existing ontologies. This feature is the architectural key to achieving componentization and isolation within the scientific vocabulary layer.

Componentization through Application Ontologies: Rather than maintaining a single monolithic “AbbVie Vocabulary,” the architecture uses CENtree to host the scientific facets of the Core Unifying Ontology as a standalone, read-only anchor. Separate Application Ontologies are created for specific domains—Neuroscience, Oncology, Clinical Trials, Immunology. Each domain ontology imports only the specific branch of the core taxonomy it needs. This prevents a scientist in the Clinical domain from being cluttered by manufacturing-specific terms from Ludwigshafen, and prevents a Ludwigshafen formulation scientist from navigating thousands of irrelevant clinical endpoint definitions.

The Proxy Pattern — Insulating from External Ontology Drift: Public ontologies like MeSH, SNOMED CT, and Gene Ontology release updates every few months. If business logic maps directly to these external vocabularies, the system requires constant updates every time an external authority restructures its hierarchy. The architecture addresses this through a proxy pattern: domain ontologies map to the stable Core Unifying Ontology first, then CENtree’s diffing and versioning tools reconcile the Core with public updates on a governed cadence. Domain-specific applications—including the AI tools built for the Semantic Gateway—are insulated because they only ever reference the stable Core, not the volatile external vocabularies directly.

Local Curation with Global Alignment: CENtree's multi-user permission system enables a "local curation, global alignment" model. Scientists at the Ludwigshafen site act as Editors for their specific Neuroscience sub-domain ontology. They can add proprietary terms (such as Melt Extrusion formulation parameters or site-specific analytical method descriptors) to their local domain without those terms ever polluting the global Core unless explicitly promoted through the governance review process. This empowers the scientific community to maintain their own source of truth in a tool designed for their vocabulary while remaining anchored to the top-down Core.

Layer	Component	CENtree Implementation	Benefit
Top	Core Unifying Ontology (scientific facets)	Master Ontology (Global Read-Only)	Enterprise-wide consistency; insulated from domain churn
Middle	Domain Ontologies	Application Ontologies (Imported Subsets)	Domain-specific focus; no cross-domain clutter
Bottom	Local Vocabularies	Proprietary Space (Collaborative Editing)	Rapid, local scientific curation with global anchoring

Level 4: Bottom-Up Source Schemas and Neo4j Graph Topology — The Digital Mirror

The bottom-up tier consists of the physical data schemas from 200+ internal and external sources, plus the Neo4j labeled property graph topology that materializes the ARK knowledge graph. These structures are inherently heterogeneous: source databases use proprietary column names, legacy codes, and inconsistent entity representations, while the Neo4j graph uses node labels, relationship types, and property keys that must conform to graph query performance requirements. The bottom-up tier is where the “disparate strings” live before they are resolved into “unified biological things.” Neo4j acts as the Digital Mirror—the knowledge graph of every physical and digital asset—reflecting the operational reality of AbbVie’s data landscape.

Insulation from Change: When a database schema changes or a piece of equipment in Ludwigshafen is replaced, only the Neo4j node and its specific mapping ontology need to update. The top-down Business Domain ontologies (Level 2) and the Core Unifying Ontology (Level 1) remain untouched. This directional insulation is critical: bottom-up changes propagate only as far as the mapping layer, never upward into the governance hierarchy.

Integration of Third-Party Data: By treating third-party ontologies (such as Tellic’s relationship extraction outputs, ChEMBL structures, and PubMed metadata) as bottom-up assets rather than embedding them in the top-down governance hierarchy, the architecture avoids hard-coding external standards into internal business logic. Third-party data enters through the MBSE-governed ingestion pipeline, receives mapping assertions that bind it to the domain ontologies, and is materialized in Neo4j with full provenance—but the external vocabulary structures themselves never contaminate the Core.

Source Schema Profiling: During Phase 1 ingestion, every source undergoes MBSE-governed schema profiling via Modak Nabu. This process inventories all tables, columns, data types, value distributions, and implicit entity references within each source. The profiling output is a machine-readable source schema model that captures the bottom-up reality of what exists in each database without imposing top-down assumptions.

Neo4j Graph Schema: The Neo4j graph topology is not a free-form graph. It is governed by a formal graph schema—a set of approved node labels, relationship types, and property key constraints—that is derived from the domain ontologies via a schema generation pipeline. Each domain ontology class maps to a Neo4j node label, each object property maps to a relationship type, and each data property maps to a node property key. This generation process is deterministic and version-controlled: when a domain ontology release introduces a new concept, the graph schema generator produces the corresponding Neo4j constraint definitions, which are reviewed, tested, and deployed through a standard CI/CD pipeline.

The Mapping Governance Layer: The Fidelity Gate

The four-level governance topology produces three distinct types of mappings, each requiring its own governance surface. The ontological hierarchy (rdfs:subClassOf and owl:equivalentClass assertions between Levels 1–3) lives in the RDF/OWL layer. The materialized result (the multi-label stack on each Neo4j node) lives in the LPG. But the actual binding assertions—“Column X in Source Y represents instances of Concept Z in Domain Ontology D”—are meta-assertions about the relationship between source schema elements and ontology concepts. They are metadata about metadata, and they require their own governed, versioned, auditable storage layer.

This Mapping Governance Layer is the most critical component of the architecture for ensuring data integrity. It functions as the Fidelity Gate between the abstract governance hierarchy and the concrete data reality—the point where top-down semantic intent meets bottom-up operational truth, and where the architecture either maintains or loses its claim to being an Architecture of Truth.

Domain-Specific AI Mapping Agents

The mapping process uses domain-specific AI agents that act as automated translators between the abstract (Level 2: “Clinical Trial”) and the concrete (Neo4j: “Table_X_in_Oracle_DB”). These agents are specialized by domain—an Oncology Mapping Agent has different entity recognition patterns and confidence calibration than a Manufacturing Mapping Agent—reflecting the fact that the linguistic and structural patterns that indicate semantic alignment vary dramatically across scientific and operational domains.

SciBite TERMite performs automated candidate matching by running NER against source metadata (column names, value samples, data dictionaries) to propose candidate ontology alignments. These candidates are then reviewed and validated by knowledge engineers using an AI-assisted mapping dashboard that presents confidence-ranked suggestions alongside the source schema profile. The MBSE traceability model guarantees completeness: every source schema element must be mapped, explicitly excluded, or flagged for remediation before the source is promoted to production ingestion. This prevents unmapped columns from silently entering the data lake as orphaned, semantically opaque artifacts.

Fidelity Thresholds and Human-in-the-Loop Integrity

To prevent the backfire of erroneous data, the architecture implements a high threshold value (0.9) for NLP-derived triples. The fidelity threshold ensures that automated mappings are not simply letting an LLM guess at alignments—it directly addresses the hallucination and data integrity concerns that are paramount in pharmaceutical R&D. Insights from the GNBR-Nexus evaluation confirm that while average scores across broad relationship classes can be modest, applying high thresholds to specific classes like “inhibits” yields high-fidelity triples not reported in structured datasets. This rigor ensures that the Knowledge Graph remains an Architecture of Truth rather than an automated producer of misleading drug discovery leads.

Below the 0.9 threshold, mapping candidates enter a human-in-the-loop review queue where knowledge engineers and domain scientists evaluate them for promotion. This hybrid approach—automated candidate generation with human validation—preserves the speed benefits of AI-driven mapping while maintaining the scientific integrity that GxP compliance demands. The fidelity threshold is not a single global setting; it is calibrated per relationship class and per domain, reflecting the fact that some semantic relationships (like “inhibits”) are expressed with higher linguistic precision than others (like “associated with”) in the biomedical literature.

Mapping Storage Architecture

The mapping assertions themselves are stored as versioned, auditable artifacts in a dedicated Mapping Registry that constitutes a distinct governance surface—separate from the ontology governance in Graph Studio, separate from the vocabulary management in CENtree, and separate from the graph schema management in Neo4j. The Mapping Registry is the fourth artifact type in the governance architecture, positioned between the scientific domain ontologies (Level 3) and the bottom-up source schemas (Level 4).

The architecture uses a hybrid storage approach. Altair Graph Studio's semantic data fabric layer hosts the formal mapping assertions as governed semantic artifacts, expressed using R2RML-compatible mapping patterns and SSSOM (Simple Standard for Sharing Ontological Mappings) where applicable. Graph Studio supports both RDF and LPG targets, handles versioning and access control natively, and provides the mapping lifecycle management the architecture requires: every mapping follows a governed lifecycle (draft, review, approved, deprecated), every mapping revision triggers downstream impact analysis, and no mapping can be deprecated while active Neo4j nodes still reference it.

The Neo4j LPG carries pointers back to these mappings—the CENtree concept ID, the source identifier, the ingestion timestamp, and the mapping version reference—just enough to enable lineage tracing without embedding the full mapping governance apparatus in the graph itself. The RDF/OWL layer carries the ontological axioms that the mappings reference—the class hierarchy, the property definitions, the constraint models—without being polluted by operational ETL metadata. The Mapping Registry bridges between them.

Ownership and Governance: The Mapping Registry is owned by the KIA team with domain-specific delegation to embedded knowledge engineers in each therapeutic area. Every mapping must be approved before source data flows into the graph. Every mapping revision triggers automated impact analysis that identifies all affected Neo4j nodes. No mapping can be deprecated while active nodes still reference it. This governance rigor ensures that the instructions for how top-down and bottom-up converge are themselves governed artifacts, not ad-hoc decisions buried in ETL scripts.

Top-Down to Bottom-Up: Graph Schema Projection

Once domain ontologies are published from CENtree and validated against the Core Unifying Ontology in Graph Studio, the graph schema projection pipeline translates the OWL/SKOS structures into the Neo4j labeled property graph topology. This projection follows deterministic transformation rules that ensure consistent, predictable graph structures.

- **Node Label Derivation:** Each domain ontology class flagged as “materializable” generates a corresponding Neo4j node label. The Core 9 upper classes are also materialized as supertype labels, enabling multi-label inheritance in the graph. For example, a node representing the molecule ABBV-951 carries both the domain label :TherapeuticCandidate and the upper ontology label :Product, allowing Cypher queries to operate at any level of abstraction.
- **Relationship Type Derivation:** Each object property in the domain ontology generates a Neo4j relationship type. The 80+ relationship types in ARK (inhibits, agonism, upregulates, ASSOCIATED_WITH_ADVERSE_EVENT, etc.) are all derived from formal ontology property definitions, not ad-hoc graph modeling decisions. Domain and range constraints from the ontology are enforced as Neo4j constraints, preventing semantically invalid edges from being created.
- **Property Key Derivation:** Data properties from the ontology generate Neo4j property keys with enforced data type constraints. Every node carries mandatory provenance properties—source identifier, ingestion timestamp, confidence score, CENtree concept ID, and mapping version reference—ensuring full traceability from graph node back through the domain ontology to the originating source schema element.
- **Polymorphic Label Inheritance:** Neo4j’s multi-label capability is exploited to materialize the full ontological hierarchy on every node. A gene target node simultaneously carries labels from all levels: the domain-specific label (e.g., :KinaseTarget), the broader domain class (e.g., :BiologicalTarget), and the Core 9 supertype (e.g., :Asset). This enables the Context Engine to resolve queries at the appropriate level of specificity—a scientist querying “all kinase targets” retrieves a different scope than a strategist querying “all assets”—while both queries traverse the same underlying graph.

Continuous Reconciliation and Drift Detection

Mapping is not a one-time activity. As source schemas evolve, as new data sources are onboarded, and as domain ontologies are refined, the alignment between levels can drift. ARCH addresses this through continuous reconciliation mechanisms.

- **Schema Drift Detection:** Automated monitors compare the current state of each source schema against its last-profiled baseline. When columns are added, removed, or changed in a source system, the drift detector flags the affected mapping assertions in the Mapping Registry for review. This prevents silent data quality degradation where a source schema change causes downstream entities to be misclassified or lost.
- **Ontology-Graph Consistency Validation:** After every domain ontology release from CENtree, an automated validation suite compares the published ontology against the live Neo4j graph schema. The validator checks for orphaned node labels (labels in the graph that no longer have ontology backing), missing materializations (new ontology concepts that have not yet been projected into the graph), constraint violations (edges that violate updated domain or range restrictions), and label hierarchy inconsistencies (nodes whose multi-label sets do not correctly reflect the current ontology subsumption hierarchy).

- **Concept ID Anchoring:** Every node in the Neo4j graph carries a CENtree concept ID as a mandatory property. This concept ID is the immutable anchor that binds the physical graph node to its position in the ontological hierarchy. When a domain ontology concept is deprecated or merged, the concept ID anchoring ensures that all affected graph nodes are identified and remediated. The concept ID also serves as the join key between the graph and the FAIR metadata catalog, enabling end-to-end lineage tracing from Cypher query result back through the graph node, through the domain ontology, through the Mapping Registry, to the specific column and row in the originating source system.

Cross-Site Semantic Bridging: The US–Germany Integration Architecture

The federated governance topology directly addresses the semantic silos that exist between AbbVie’s US-based R&D operations (Lake County) and the German scientific heritage at the Ludwigshafen site. Scientists in Ludwigshafen (Neuroscience and Melt Extrusion focus) and Lake County (broad R&D portfolio) often have conflicting definitions for the same data tags. A single, centralized vocabulary forces one side to lose its scientific precision to accommodate the other—creating what amounts to semantic overload that degrades both sides’ ability to work with their own data.

The componentized architecture resolves this through domain-specific veneers that preserve local scientific rigor while providing enterprise-wide translation through the Core Unifying Ontology.

The Isolate–Map–Automate Pattern

Step 1 — Isolate: A dedicated Ludwigshafen Neuroscience Component is created as a standalone Application Ontology in CENtree. This component captures the specific scientific vocabulary of the German site—unique terms for Melt Extrusion formulation parameters, Alzheimer’s deep-science terminology, site-specific analytical method descriptors, and the 130 years of German scientific rigor embedded in Ludwigshafen’s research archives. German scientists maintain their specific terminology and precision without compromise.

Step 2 — Map: Local terms from the Ludwigshafen component are linked to the stable Core Unifying Ontology through formal mapping assertions managed in the Mapping Registry. This creates a Universal Translator that does not replace German terms—it defines how they translate into Global AbbVie concepts for the US teams. The Core Unifying Ontology acts as the bridge layer: it does not clutter the US teams’ daily workflow with German-specific terminology, and it does not force the German teams to abandon their precise scientific language.

Step 3 — Automate: The componentized mapping logic triggers automated cross-site queries. When a German breakthrough in neuroscience (expressed in Ludwigshafen-specific vocabulary) matches a US clinical need (expressed in Lake County clinical trial terminology), the Semantic Gateway identifies the match through the Core Unifying Ontology mapping without requiring manual data hunting. The Context Engine’s 5 Pillars of Relevancy operate across both vocabularies simultaneously because both map to the same Core—the bridge is structural, not ad-hoc.

Step	Action	Outcome
1. Isolate	Create standalone Ludwigshafen Neuroscience Component in CENtree	German scientists maintain specific scientific rigor and terminology without compromise
2. Map	Link local terms to Core Unifying Ontology via Mapping Registry	Universal Translator created; US teams see Global AbbVie concepts without German-specific clutter
3. Automate	Componentized logic triggers automated cross-site Semantic Gateway queries	Instantly identifies when German breakthroughs match US clinical needs without manual data hunting

Federated Governance Topology Summary

Level	Component	Platform	Governance Cadence	Isolation Boundary
Level 1	Core Unifying Ontology (Core 9)	Altair Graph Studio (OWL 2, SHACL)	Quarterly; cross-functional change control	Insulated from scientific churn and schema changes
Level 2	Business Domain Ontologies	Altair Graph Studio; published to CENtree	Monthly; KIA team review with automated Core validation	Domain-scoped; new business units plug in without Core redesign
Level 3	Scientific Domain Ontologies	SciBite CENtree (Application Ontologies)	Monthly; scientist-curated with global alignment checkpoints	Therapeutic-area scoped; local curation without global pollution
Mapping Layer	Mapping Registry (Fidelity Gate)	Graph Studio semantic fabric + AI Agents	Continuous; automated drift detection on every ingestion cycle	Binding assertions versioned independently from both ontologies and graph
Level 4	Source Schemas + Neo4j Graph Topology	Modak Nabu (profiling); Neo4j (materialization)	Continuous; CI/CD graph schema deployment on CENtree release	Schema changes propagate only to Mapping Layer, never upward

End-to-End Lineage: From Cypher Query to Source Row

The federated mapping architecture produces complete, auditable lineage for every piece of knowledge in the ARCH graph. When a scientist executes a Cypher query and receives a result—for example, “ABBV-951 inhibits KinaseTarget-X with confidence 0.94”—the lineage chain is fully traceable. The Neo4j node carries a CENtree concept ID that links to the domain ontology concept (e.g., Oncology Target Ontology: KinaseTarget-X). The domain ontology concept has a formal subsumption path through Level 3 (scientific domain), Level 2 (business domain), to Level 1 (Core Unifying Ontology): KinaseTarget-X → BiologicalTarget → Asset. The node’s provenance properties identify the source (e.g., Tellic external extraction, PubMed article PMID:12345678). The Mapping Registry records which source schema element was mapped to this concept, which AI Agent proposed the mapping, what the confidence score was, and which human reviewer approved it. And the MBSE traceability model confirms that the mapping was complete and validated. This lineage is what transforms ARCH from a data retrieval system into a GxP-compliant Architecture of Truth where every assertion can be defended under regulatory scrutiny.

Ontological Framework: Core 9 Mapping to ARCH

The ARCH platform is governed by a set of nine core ontologies—Process, Service, Asset, Product, Party, Location, Time, Event, and Issue—that provide the stable architectural lineage ensuring every data point remains traceable to an enterprise-wide Unifying Ontology. This upper-level semantic layer governs disparate, bottom-up data schemas while ensuring polymorphic meaning across different business domains.

Semantic Blueprint: Core 9 to Dossier Mapping

Core Ontology	ARCH Dossier Application	Strategic Value
Product	The therapeutic candidate or molecule being investigated	Ensures the Digital Twin of a molecule is consistent across all R&D silos (Lars Greiffenberg)
Asset	Internal genomics data, chemical DELs, and Tellic's 1B+ biomedical relationships	Provides clear inventory of Integrated Knowledge assets for GxP auditing (Michael Kaltschmidt)
Process	Drug discovery lifecycle from hypothesis generation to clinical trial execution	Maps LAM workflows to specific R&D milestones (Brian Martin)
Party	Scientists querying the graph, practitioners in the oncology portal, and patients	Drives User Enablement by tailoring UX to each party's vocabulary (Elfi Stach)
Location	Therapeutic areas (Oncology, Immunology) and global research sites	Harmonizes data originating from different global sites (Lake County vs. Ludwigshafen)
Time	Phased execution timeline and trial durations	Anchors the 24-month roadmap to measurable temporal milestones
Event	Data ingestion triggers, milestone achievements, and patient outcome observations	Tracks provenance by recording exactly when and how knowledge entered ARCH
Issue	Integration debt, AI hallucinations, and regulatory compliance risks	Uses the ontology to flag and track hallucination-free status for AI outputs
Service	AI-driven insight delivery, RAG frameworks, and the Gaia Platform	Transitions the ontology from a static database to an active research accelerator

Polymorphic Meaning and Integration Debt Resolution

The Core 9 mapping demonstrates that as a Product (molecule) undergoes transformation across different systems—from a string in a published paper to a thing in a chemical library—it maintains its identity via the Core Product Ontology. This recursive identity preservation is the key to solving integration debt: ARCH is not another silo but a system where every dossier view is a business-specific projection of the enterprise-wide core ontology.

Context Engine: Operational Demonstration

The following walkthrough illustrates how the Context Engine transforms a simple user query into a precise, ground-truth Cypher query, utilizing the 5 Pillars of Relevancy and Core 9 ontological mapping.

Step 1: Raw Input

User Query: "What is the latest on drug safety for our new oncology target?"

Step 2: Context Engine Analysis (5 Pillars)

Pillar	Extracted Ontological Metadata	Source	Core 9
Who	Senior Research Fellow (Lars Greiffenberg persona). Permissions: GxP / Internal R&D.	Party	Party
What	Target: ABBV-951 (mapped from "new oncology target"). Concept: Adverse Events & Toxicology.	Product / Asset	Product / Asset
Where	Domain: Oncology R&D. Site: Lake County / ARCH Production Graph.	Location	Location
When	Scope: Post-January 2026 (filtered for "latest" clinical trial uploads).	Time / Event	Time / Event
Why	Intent: Target/Safety Dossier authoring for regulatory submission.	Process / Issue	Process / Issue

Step 3: Semantic Gateway (Cypher Generation)

The Enterprise Prompt Library uses the enriched metadata to generate a precise Cypher query:

```
// Generated by Context Engine for Persona: R&D_Scientist
MATCH (p:Product {name: 'ABBV-951'})-[:HAS_TARGET]->(t:BiologicalTarget)
MATCH (t)-[r:ASSOCIATED_WITH_ADVERSE_EVENT]-(ae:Issue)
WHERE r.timestamp > '2026-01-01' AND r.source = 'Internal_Clinical_Trial'
RETURN p.name, t.name, ae.description, r.severity, r.timestamp ORDER BY
r.timestamp DESC LIMIT 5
```

Step 4: Verified Articulation

Instead of a generic LLM response, the UI renders a Safety Dossier view with grounded, traceable results:

> *For Target ABBV-951: 3 new Grade 2 neutropenia events were recorded in the ARCH Hub since your last login (Feb 2026). These events correlate with Tellic's external findings for similar protein kinase inhibitors.*

Every element of this response is traceable: the Who (persona and permissions), What (specific molecule and adverse event type), Where (production graph and oncology domain), When (post-January 2026 filter), and Why (dossier authoring intent) are all auditable, satisfying GxP compliance requirements.

Stakeholder Value Delivery

The Patient

Knowledge integration accelerates the delivery of first-in-kind medicines, which now represent 75% of AbbVie's pipeline. ARCH has already demonstrated value in drug repurposing, identifying potential therapies for complex diseases and COVID-19 by flagging existing molecules with high-fidelity potential. Patient Digital Twins further personalize therapeutic approaches by simulating individual patient responses before clinical intervention, while the Rare Hopes UX ensures that patients with rare diseases gain access to graph-derived insights that would otherwise remain locked in research literature.

The Practitioner and Scientist

The framework empowers scientists through the Target and Safety Dossiers, which aggregate 18 distinct views of a target into one interface. Researchers receive a comprehensive view of pathways, variants, and phenotypes in minutes, pivoting from data gathering to innovative experimentation. The AbbVie Science Oncology Platform extends this value to external practitioners, delivering curated research updates directly to oncologists. During Phase 3, wearable-derived DHT endpoints provide objective, real-world patient outcome data that complements traditional clinical measures.

The AbbVie Enterprise

The ARCH platform catalyzes the immunology encore, bolstering expansion into neuroscience and obesity as primary growth drivers. Operationally, the platform prevents duplication of expensive datasets and ensures supply resilience across future product cycles. The unified knowledge layer provides quantifiable ROI through reduced trial redundancy, compressed discovery timelines, and lower late-stage attrition rates.

Cultural Transformation: The AbbVie Convergence Journal

Value realization is anchored in cultural change. The AbbVie Convergence Journal—a quarterly, internal, peer-reviewed publication—showcases real-world use cases where scientists build their own products atop the ARCH ecosystem. This transformational commitment ensures the platform becomes a centerpiece of AbbVie's converged scientific culture.

Data Governance & Quality Standards

FAIR Principles as Foundational Mandate

All data entering the ARCH ecosystem must satisfy strict adherence to FAIR Principles across all four dimensions. These are not aspirational guidelines but unyielding operational mandates enforced through automated quality gating.

- **Findability:** Every data asset receives persistent, unique identifiers and is registered in a searchable metadata catalog. Rich descriptive metadata enables discovery across the 200+ connected sources.
- **Accessibility:** Data is retrievable through standardized, open protocols with clearly defined authentication and authorization mechanisms. Access policies are persona-aware, governed by the Context Engine's Party ontology.
- **Interoperability:** All data uses formal, shared, broadly applicable vocabularies managed through SciBite CENtree. Ontological mapping ensures that concepts are semantically interoperable across therapeutic domains and organizational boundaries.
- **Reusability:** Data assets carry clear provenance metadata, usage licensing, and domain-relevant community standards. Every asset is documented for secondary use, enabling risk-based data democratization across the R&D enterprise.

MBSE as Core Ingestion Methodology

Model-Based Systems Engineering serves as the primary methodology governing all data ingestion and integration workflows within ARCH. Rather than ad-hoc ETL development, every integration pipeline is designed, validated, and documented as a formal systems engineering model. MBSE guarantees zero data loss during migration from legacy systems by providing complete traceability from source schema to target ontology, structural validation of mapping completeness before execution, automated regression testing against known data integrity benchmarks, and prevention of integration debt through rigorous architectural governance. This approach ensures that the ARCH data foundation is structurally sound from inception, avoiding the accumulation of technical debt that plagues conventional data lake implementations.

Service Level Agreements (SLAs)

Unyielding SLAs govern data quality, availability, and freshness across all ARCH data assets. These SLAs are monitored continuously through automated dashboards and are integrated into the FAIR quality gateway. Non-compliant data assets are quarantined until remediation is complete, preventing erroneous or stale data from contaminating the knowledge graph.

Strategic Risk Mitigation

The ARCH platform serves as a primary mitigation strategy for the four core strategic risks facing AbbVie:

Debt & Capital Risk

By increasing R&D efficiency and preventing late-stage clinical failure, ARCH maximizes the return on R&D investment to balance high debt levels. Digital Twin simulations further reduce capital risk by optimizing trial designs and reducing redundant experiments before capital is committed.

Execution Risk

AbbVie is investing $380 million in new AI-powered manufacturing facilities in North Chicago. ARCH acts as the pipeline feeder for these sites, ensuring they are utilized for high-quality, validated medicines. The 24-month phased roadmap with explicit milestones provides continuous visibility into execution progress.

Regulatory Pressure

A traceable knowledge graph provides the evidence and provenance required to satisfy intensified pricing and regulatory scrutiny. The Context Engine’s full auditability—capturing Who, What, Where, When, and Why for every query and response—delivers GxP-compliant documentation automatically.

Margin Stability

By shortening discovery timelines and driving the immunology encore in oncology and obesity, the framework protects margins against competitive pressures. The integration of DHT and wearable data in Phase 3 generates real-world evidence that supports value-based pricing and market access strategies.

Leadership Engagement Strategy

The ARCH solution architecture directly addresses the strategic priorities of key stakeholders across the organization:

- **Brian Martin:** The Context Engine demonstration proves that the Semantic Gateway is not a prompt wrapper but a systematic architecture guaranteeing RAG Grounding by forcing the LLM to speak the language of the graph (Cypher) rather than guessing. The Fidelity Gate architecture—with domain-specific AI Mapping Agents enforcing a 0.9 confidence threshold and human-in-the-loop validation—directly addresses the hallucination and data integrity concerns critical to AI-driven drug discovery. The LAM evolution in Phase 3 maps action model workflows directly to R&D milestones, with the Mapping Registry providing full auditability for every AI-generated insight.
- **Michael Kaltschmidt:** The FAIR-compliant, MBSE-governed ingestion pipeline ensures GxP-readiness. The four-level federated governance topology—with the Mapping Registry providing versioned, auditable binding assertions between source schemas and ontology concepts—creates end-to-end lineage from Cypher query result to source row. Every query is traceable through the graph node, through the domain ontology, through the Mapping Registry, to the specific column and row in the originating source system. The Core 9 mapping demonstrates that ARCH resolves integration debt through recursive identity preservation, not silo construction.
- **Lars Greiffenberg:** The Product ontology ensures the Digital Twin of a molecule is consistent across all R&D silos. Phase 2’s Patient Digital Twins extend this concept to patient-level simulation, creating a comprehensive twin ecosystem spanning molecules and patients.
- **Elfi Stach:** The Party ontology drives User Enablement by tailoring UX to each persona’s vocabulary. The Rare Hopes UX and AbbVie Science Oncology Platform demonstrate concrete platform extensions reaching patients and practitioners beyond internal R&D.

This Solution Architecture is the engine required to transform the biopharmaceutical enterprise, unlocking the information that makes cures possible.

Appendix A

Strategic Technology Assessment: Mendix + Altair for ARCH Acceleration

Leveraging the Siemens Xcelerator Platform for Component-Based Knowledge Application Delivery and Semantic Governance

Overview

This assessment evaluates the strategic fit of Mendix (low-code application platform) and Altair Studio (data science, knowledge graph, and simulation platform) as complementary technology layers for the AbbVie R&D Convergence Hub (ARCH). The central finding is that both platforms are now unified under Siemens Xcelerator following the $10 billion Altair acquisition completed in March 2025, creating what Siemens describes as the world's most complete AI-powered portfolio of industrial software for simulation, HPC, data science, and artificial intelligence.

This convergence creates a unique architectural opportunity. Mendix addresses the application delivery bottleneck at ARCH Layer 4, enabling component-based rollout of dossier views, the Rare Hopes UX, and the AbbVie Science Oncology Platform through low-code microapps governed by the same semantic ontology layer. Altair's portfolio—particularly Graph Studio (formerly Anzo by Cambridge Semantics), AI Studio (formerly RapidMiner), and the HyperWorks simulation engine—addresses critical gaps in ontology management tooling, automated semantic mapping, Patient Digital Twin simulation, and Graph RAG grounding.

Most importantly, this architecture directly mirrors proven enterprise semantic governance patterns: an upper ontology governing federated domain vocabularies, mapped through automated pipelines to bottom-up data schemas, with composable applications consuming the knowledge layer through a context-aware semantic gateway. The Siemens Xcelerator integration provides the industrial-grade platform substrate that makes these patterns executable at pharma R&D scale.

The Siemens Xcelerator Convergence

The strategic significance of this recommendation rests on a platform convergence that most observers have not yet fully internalized. As of March 2025, Siemens owns both Mendix and Altair, and both are being integrated into the Siemens Xcelerator open digital business platform. This is not a loose partnership—it is a single-vendor portfolio with native interoperability commitments.

What Siemens Now Controls Under One Roof

- **Mendix:** Gartner Magic Quadrant Leader for Enterprise Low-Code Application Platforms (2025). Visual drag-and-drop development, embedded ML Kit for in-app model execution, native AWS Bedrock and OpenAI integration, Git-based version control, Maia AI co-developer, and cloud-native containerized deployment.
- **Altair Graph Studio (formerly Anzo by Cambridge Semantics):** Enterprise-scale knowledge graph toolset that applies a semantic, graph-based data fabric layer over diverse data sources. Built on RDF/OWL with native ontology modeling, SPARQL/SPARQL* support, Graph Lakehouse MPP engine, entity resolution, Graph RAG capabilities, and a conversational CoPilot interface. Supports both RDF triple stores and labeled property graph models.
- **Altair AI Studio (formerly RapidMiner):** Gartner Magic Quadrant Leader for Data Science and Machine Learning Platforms (2025). No-code drag-and-drop ML workflow designer, AutoML, generative AI support, data fabric integration, and production-ready model deployment.
- **Altair HyperWorks:** The industry's leading design and simulation platform with multiphysics simulation, digital twin capabilities, AI-powered engineering optimization, and Twin Activate for system-level modeling and code generation.
- **Siemens Xcelerator Ecosystem:** Teamcenter (PLM), Opcenter (MES), Insights Hub (IoT), Polarion (requirements/ALM)—all accessible as data sources and integration targets for Mendix applications.

The Altair + Mendix integration is already being demonstrated in production. At the 2025 ATCx conference, Siemens presented jointly on the Evora Factory Management application—a Mendix low-code app consuming Altair analytics to deliver a 360-degree operational view of factory performance. This is the exact architectural pattern ARCH needs for its Layer 4 dossier applications.

Mapping to the ARCH Architecture

The ARCH solution architecture is organized as a four-layer stack: Data Foundation (Layer 1), Semantic Harmonization (Layer 2), AI and Prompt Engineering (Layer 3), and User Interface and Application Delivery (Layer 4). The Mendix + Altair combination addresses specific technical gaps and acceleration opportunities at every layer.

Layer 1: Data Foundation — Altair Graph Studio for Semantic Data Fabric

The current ARCH Layer 1 relies on Modak Nabu for data movement and SciBite TERMite/CENtree for NER and ontology management. Graph Studio does not replace these tools but introduces a complementary semantic data fabric layer that sits above the raw data lake and below the Neo4j knowledge graph.

- **Automated Source Schema Profiling:** Graph Studio's data onboarding pipelines automatically catalog and connect structured and unstructured data sources into knowledge graph structures. This directly accelerates the MBSE-governed source schema profiling described in the Semantic Governance Architecture section, providing machine-readable schema discovery that feeds the Stage A semantic alignment mapping.
- **Entity Resolution at Scale:** Graph Studio includes built-in entity resolution, deduplication, and data cleaning capabilities—the exact operations performed by the Reveal-Reframe-Resist model. Graph Studio's semantic layer can enforce the same context-driven disambiguation rules that prevent NLP from confusing COX-1 (Prostaglandin G/H synthase 1) with Cytochrome c oxidase.
- **FAIR Metadata Automation:** By applying a semantic layer over all data sources, Graph Studio creates the machine-readable metadata catalog that FAIR Findability and Reusability require. Every data asset's semantic meaning, provenance, and lineage are captured in the graph fabric automatically.

Layer 2: Semantic Harmonization — Graph Studio as Ontology Governance Engine

This is where the alignment with the ARCH Semantic Governance Architecture becomes most direct. Graph Studio is built on the same W3C standards stack—RDF, OWL 2, RDFS, SKOS—that governs the three-tier ontology model in the ARCH Semantic Governance Architecture.

- **Native OWL 2 Ontology Modeling:** Graph Studio's Graph Lakehouse includes built-in modeling tools for creating, editing, and importing models based on RDF and OWL. This provides an enterprise-grade authoring environment for the Tier 1 Upper Ontology (Core 9) that is currently specified as requiring Protégé or TopBraid-class tooling. Graph Studio offers the same formal semantics with native versioning, access control, and collaborative development.
- **Dual Graph Model Support:** Critically, Graph Studio supports both the W3C RDF data model and the labeled property graph model via RDF* and SPARQL*. This means it can natively bridge between the OWL-based upper and domain ontologies (Tiers 1–2) and the Neo4j labeled property graph topology (Tier 3) without requiring a separate schema

projection pipeline. The same platform that authors the ontology can generate and validate the graph schema.

- **Semantic Layer Over Neo4j:** Graph Studio can operate as a semantic data fabric layer over the existing Neo4j ARK graph, adding formal ontological definitions on top of the property graph without requiring a migration away from Neo4j. This preserves AbbVie's investment in Neo4j while adding the governance tooling that the architecture requires.
- **Graph RAG Integration:** Graph Studio's Graph RAG capabilities directly support the Federated RAG 2.0 architecture in ARCH Layer 3. By constructing vector embeddings from the ontology-enriched graph, Graph Studio provides the grounding context that constrains LLM responses to verified semantic data—the exact mechanism described in the Semantic Gateway.

Layer 3: AI and Prompt Engineering — Altair AI Studio for ML Pipelines

Altair AI Studio addresses the machine learning operational gap in the current ARCH architecture. While ARCH defines the AI layer in terms of the AWS Bedrock Router and Enterprise Prompt Library, it lacks a specified platform for the ML model lifecycle that underlies Digital Twin simulation, predictive safety analytics, and NER model training.

- **Digital Twin Model Development:** AI Studio's drag-and-drop workflow designer enables domain scientists to build, train, and validate Patient Digital Twin predictive models without requiring data science team bottlenecks. AutoML capabilities accelerate model selection and hyperparameter optimization for clinical outcome prediction.
- **NER Model Enhancement:** AI Studio can serve as the training and evaluation environment for custom NER models that supplement SciBite TERMite, particularly for AbbVie-specific entity types not covered by public biomedical ontologies.
- **Embedded ML via Mendix:** The Mendix ML Kit allows trained AI Studio models to be embedded directly into application runtimes, eliminating the latency and complexity of API-based model serving. A Dossier application can run a safety signal classifier locally rather than round-tripping to a centralized inference endpoint.

Layer 4: User Interface — Mendix for Component-Based Application Delivery

This is where Mendix delivers the most immediate timeline acceleration. The current ARCH Layer 4 specifies TXI Digital and the Unity design system for application delivery, producing six distinct platforms: ARCH Search, Target Dossier, Safety Dossier, AbbVie Science Oncology Platform, Rare Hopes UX, and Ask ARCH. Each of these is a custom-developed application. Mendix transforms this from a serial development pipeline into a composable, component-based rollout.

- **Component-Based Microapp Architecture:** Each dossier view, each portal, and each specialized UX becomes a Mendix microapp—independently deployable, independently scalable, and independently evolvable. The 18 distinct views in the Target and Safety Dossiers can be developed as reusable components that are composed into different application contexts without code duplication.
- **Scientist-Authored Applications:** Mendix's low-code visual development environment enables domain scientists to build their own applications on top of the ARCH knowledge layer. This directly supports the cultural transformation described in the AbbVie

Convergence Journal—scientists building products atop the ARCH ecosystem—without requiring each scientist to become a full-stack developer.

- **Persona-Driven UX Delivery:** The Context Engine's Party ontology maps directly to Mendix's role-based access model. The same underlying knowledge graph query can render different Mendix application views for the Scientist (Target Dossier), Practitioner (Oncology Platform), and Patient (Rare Hopes UX) personas—each consuming the same Cypher query results through persona-tailored Mendix interfaces.
- **10x Development Velocity:** Mendix consistently claims 10x faster delivery than traditional development for enterprise applications. Even at a conservative 5x multiplier, this compresses the Phase 2 application deployment timeline significantly, potentially enabling parallel rather than sequential rollout of the six Layer 4 platforms.

Alignment with Proven Enterprise Semantic Governance Patterns

The ARCH architecture already incorporates foundational enterprise semantic governance patterns: a Core 9 upper ontology governing federated domain vocabularies, a three-tier mapping architecture binding top-down semantics to bottom-up data realities, and a Context Engine that resolves polymorphic meaning at the point of consumption. The Mendix + Altair combination provides the industrial-grade tooling substrate that makes these patterns operationally executable rather than architecturally aspirational.

Pattern 1: Upper Ontology Governance via Graph Studio

In large-scale enterprise deployments, the upper ontology layer serves as the stable semantic ceiling governing tens of thousands of data assets. In the ARCH context, Graph Studio's OWL 2 modeling environment with formal versioning, access control, and SHACL constraint validation provides the same governance rigor for the Core 9. The quarterly change-control cadence, cross-functional review process, and stability mandate translate directly—Graph Studio simply provides enterprise-grade tooling for what the architecture already requires.

Pattern 2: Federated Domain Vocabulary Management via CENtree + Graph Studio

Proven federated ontology architectures use a domain vocabulary layer where domain-specific vocabularies specialize the upper ontology through formal subsumption assertions. In ARCH, SciBite CENtree already serves this function for biomedical domain vocabularies. Graph Studio complements CENtree by providing the semantic data fabric that connects domain vocabularies to actual source data schemas—the Stage A Semantic Alignment Mapping. CENtree manages the vocabularies; Graph Studio manages the mapping between those vocabularies and the physical data reality.

Pattern 3: Automated Schema Mapping and Drift Detection

The AI-assisted mapping dashboard described in the ARCH Semantic Governance Architecture—where TERMite proposes candidate alignments and knowledge engineers validate them—maps directly to Graph Studio's data onboarding and transformation pipelines. Graph Studio's ability to automatically discover, catalog, and propose semantic mappings for new data sources accelerates the MBSE-governed ingestion process while preserving the human-in-the-loop validation that ensures mapping accuracy.

Pattern 4: Component-Based Application Composability via Mendix

In mature semantic governance implementations, the ontology layer enables composable application delivery—different business domains consume the same underlying knowledge layer through domain-specific application views. Mendix provides the low-code substrate for the same pattern at AbbVie. Each ARCH dossier, portal, or UX is a composable Mendix microapp that consumes Neo4j/Graph Studio knowledge through the Semantic Gateway. New application views can be assembled from reusable components without rebuilding the knowledge layer, and the ontology's polymorphic meaning resolution ensures that each view gets contextually appropriate results.

Pattern 5: Recursive Learning via Gaia + AI Studio

The Phase 3 evolution toward the Gaia Platform—where the ontology does not just inform but executes—requires a machine learning operational backbone that the current architecture does not specify. AI Studio provides that backbone: trained models that feed predictions back into the knowledge graph, triggering ontology enrichment cycles that the Gaia Platform manages. Proven continuous learning architectures depend on this same feedback loop; AI Studio makes it operationally concrete.

Timeline Acceleration Analysis

The primary timeline impact of Mendix + Altair adoption falls on Phase 2 (Months 6–12) and Phase 3 (Months 12–24), where application delivery and advanced capabilities currently represent the critical path.

Phase 1 Impact: Moderate Acceleration

- **Graph Studio Data Onboarding:** Automates source schema profiling and candidate semantic alignment mapping, reducing the manual MBSE mapping effort for the 200+ source integration. Estimated impact: 20–30% reduction in Stage I mapping labor.
- **Entity Resolution Acceleration:** Graph Studio's built-in entity resolution capabilities complement TERMite, reducing the manual expert curation effort for the Reveal-Reframe-Resist governance model.

Phase 2 Impact: Significant Acceleration

- **Parallel Application Rollout:** With Mendix, the six Layer 4 platforms (ARCH Search, Target Dossier, Safety Dossier, Oncology Platform, Rare Hopes UX, Ask ARCH) can be developed in parallel by different fusion teams rather than sequentially by a centralized TXI Digital team. This is the single largest timeline acceleration opportunity.
- **Patient Digital Twin Prototyping:** AI Studio's AutoML and visual workflow designer enable rapid Digital Twin model prototyping, compressing the observational data integration and validation cycle. HyperWorks simulation capabilities provide physics-informed modeling that goes beyond statistical prediction.
- **Context Engine Deployment:** Graph Studio's Graph RAG capabilities can accelerate the Federated RAG 2.0 deployment by providing pre-built grounding context pipelines that integrate with the AWS Bedrock Router.

Phase 3 Impact: Capability Expansion

- **Gaia Platform Foundation:** AI Studio's MLOps capabilities provide the model lifecycle management that the Gaia recursive learning platform requires. Models trained in AI Studio, deployed through Mendix, and validated against the Graph Studio knowledge graph create the closed-loop learning architecture that Gaia describes.
- **DHT and Wearable Integration:** Mendix's native IoT integration through Siemens Insights Hub provides a pre-built data pipeline for wearable device data streams, accelerating Phase 3 DHT endpoint integration.
- **Large Action Model Development:** AI Studio's workflow automation and Mendix's business process automation capabilities provide the substrate for LAM development—multi-step R&D workflow agents that orchestrate across the knowledge graph, ML models, and application interfaces.

Technology Mapping Matrix

The following matrix maps each Siemens Xcelerator component to its specific role in the ARCH architecture, showing how it complements or extends existing ARCH technology partners.

Xcelerator Component	ARCH Layer	Function	Complements	Lifecycle Phase
Altair Graph Studio	Layers 1–2	Semantic data fabric, ontology modeling, entity resolution, Graph RAG	SciBite CENtree, Neo4j, Tellic	Phases 1–3
Altair AI Studio	Layer 3	ML model lifecycle, AutoML, Digital Twin training, NER enhancement	AWS Bedrock, SciBite TERMite	Phases 2–3
Altair HyperWorks	Layer 3	Multiphysics simulation, Patient Digital Twin modeling	Gaia Platform	Phase 3
Mendix	Layer 4	Low-code microapps, component-based dossiers, scientist-authored tools	TXI Digital / Unity	Phases 2–3
Mendix ML Kit	Layers 3–4	Embedded model inference in application runtime	AWS Bedrock Router	Phase 2
Siemens Insights Hub	Layer 1	IoT/wearable data ingestion for DHT endpoints	Modak Nabu	Phase 3

Lifecycle Rigor and Governance Implications

Beyond timeline acceleration, the Mendix + Altair combination strengthens the lifecycle governance that the ARCH Semantic Governance Architecture requires but does not fully specify in its implementation tooling.

Ontology Lifecycle Management

Graph Studio provides the complete ontology lifecycle that the three-tier architecture demands: authoring with formal OWL 2 semantics, collaborative review with version control and access permissions, automated SHACL constraint validation against the Core 9, publication to downstream consumers via APIs and export formats, and deprecation management with impact analysis across the graph. The current architecture specifies these lifecycle stages but relies on a combination of Protégé, CENtree, and manual processes. Graph Studio consolidates the governance toolchain.

Application Lifecycle Management

Mendix's built-in application lifecycle management—Agile project management, backlog management, feedback management, Git-based version control, automated testing, and CI/CD deployment—provides the same governance rigor for Layer 4 applications that the Semantic Governance Architecture provides for ontologies. Each Mendix microapp has a governed lifecycle: versioned releases, role-based access, automated quality gates, and audit trails. This prevents the Layer 4 application portfolio from accumulating the same integration debt that the ontology governance architecture is designed to prevent at the data layer.

Model Lifecycle Management

AI Studio's MLOps capabilities close the governance gap around machine learning models. Every model used in ARCH—NER classifiers, Digital Twin predictors, safety signal detectors, prompt templates—requires the same lifecycle governance as data and applications: versioning, validation against known benchmarks, staged deployment, monitoring for drift, and retirement with documented rationale. AI Studio provides this lifecycle management natively, ensuring that ML models are governed with the same rigor as the ontologies they depend on.

End-to-End Traceability

The combination of Graph Studio (data and ontology lineage), Mendix (application audit trails), and AI Studio (model provenance) creates a complete traceability chain that satisfies GxP requirements across the full ARCH stack. When a scientist receives a Safety Dossier result through a Mendix application, the lineage traces from the application version, through the model version that generated the insight, through the graph query provenance, through the ontology version that governed the mapping, to the source data row—all within governed, auditable platforms.

Risks and Considerations

Vendor Concentration

Adopting Mendix + Altair concentrates significant platform dependency on Siemens. While the Xcelerator platform is positioned as open and extensible, AbbVie should evaluate lock-in risk against the interoperability benefits. Mitigation: Graph Studio supports W3C open standards (RDF, OWL, SPARQL), Mendix exports to standard containers and APIs, and the existing Neo4j, SciBite, and AWS Bedrock investments are preserved as complementary rather than replaced.

Integration Maturity

The Altair acquisition closed in March 2025. While Mendix and Altair are already being demonstrated together (Evora Factory Management), deep product integration across the full Xcelerator stack is still maturing. AbbVie would be an early mover in applying this combined stack to pharmaceutical R&D. Mitigation: Start with a focused Phase 2 proof of concept—a single Mendix-built dossier consuming Graph Studio's semantic layer over the existing Neo4j ARK graph—before committing to full-stack adoption.

SciBite CENtree Coexistence

Graph Studio's OWL modeling capabilities partially overlap with CENtree's vocabulary management. The architecture must clearly delineate which tool governs which tier: CENtree for Tier 2 biomedical domain vocabularies (where its NER integration and life sciences vocabulary coverage are mature), Graph Studio for Tier 1 upper ontology governance and the Stage A/B/C mapping pipelines. This is a governance decision, not a technology replacement.

Cultural Adoption

Mendix's low-code promise requires genuine adoption by fusion teams—domain scientists working alongside developers. This cultural shift is already mandated by the AbbVie Convergence Journal vision but introducing a new development platform during an active 24-month roadmap adds change management complexity. Mitigation: Align Mendix adoption with the Phase 2 Rare Hopes UX deployment, where the patient-clinician focus creates natural motivation for rapid, iterative application development.

Recommendation

The Mendix + Altair combination is not just beneficial for ARCH—it is architecturally natural. The Siemens Xcelerator convergence has created a platform stack that mirrors the exact semantic governance patterns this architecture requires: formal ontology authoring with OWL 2 semantics, automated mapping between top-down ontologies and bottom-up data schemas, Graph RAG grounding for LLM integration, component-based application delivery governed by the same semantic layer, and ML lifecycle management for Digital Twin and predictive models.

The recommended adoption path is phased:

Timeline	Action	Scope	Success Criteria
Months 4–6	Proof of Concept	Single Mendix dossier view consuming Graph Studio semantic layer over existing Neo4j ARK graph	Dossier renders with full ontological lineage tracing; development velocity exceeds Unity baseline
Months 6–9	Pilot Expansion	Graph Studio deployed as Tier 1 ontology governance engine; AI Studio deployed for Digital Twin model prototyping; Mendix adopted for Rare Hopes UX	Core 9 under formal Graph Studio version control; first Digital Twin model validated; Rare Hopes UX in user testing
Months 9–12	Full Phase 2 Integration	All Layer 4 platforms developed as Mendix microapps; Graph RAG pipeline operational; CENtree–Graph Studio governance delineation finalized	Six platforms deployed in parallel; Context Engine consuming Graph Studio grounding; GxP audit trail complete
Months 12–24	Phase 3 Foundation	AI Studio MLOps for Gaia Platform; HyperWorks for Patient Digital Twin maturation; Insights Hub for DHT wearable data	Recursive learning loop operational; physics-informed Digital Twins validated; DHT endpoints integrated

This approach preserves every existing ARCH technology investment—Neo4j, SciBite, Tellic, Modak Nabu, AWS Bedrock—while adding the governed application delivery, ontology lifecycle management, and ML operational backbone that the architecture requires. The Siemens Xcelerator platform provides the integration fabric; proven enterprise semantic governance patterns provide the architectural discipline; and ARCH provides the domain-specific knowledge that makes the whole system valuable.

The question is not whether AbbVie can benefit from Mendix + Altair. The question is whether ARCH can achieve its 24-month mandate without the lifecycle rigor and composable delivery velocity that this platform combination provides.